Quilee Simeon

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Applied Scientist at Amazon FAI building LLM evaluation and data-quality systems: agentic audit pipelines, LLM-as-judge graders calibrated to STEM experts, and RL environments for coding, STEM reasoning, and agentic tool use. Background in ML and computational neuroscience (M.S., MIT); builds with coding agents every day.

EXPERIENCE

Applied Scientist · Amazon FAI (formerly AGI), Boston, MA — May 2026 – Present

- Build agent-based pipelines that automatically assess the quality of STEM expert data (math, physics, chemistry, biology, computer science), replacing much of the first-pass manual expert review.
- Develop LLM-as-a-judge graders with critical and non-critical rubric checks, calibrated against expert labels (Cohen's κ, MCC), that catch defective expert data before it reaches training.
- Build RL environments and evaluation harnesses (Harbor, Docker) spanning software engineering, STEM reasoning, and agentic tool use, graded by automated verifiers.
- Write task rubrics and guidelines with STEM domain experts and deliver quality feedback to data vendors; pipelines run on AWS (Bedrock, S3, EC2, Lambda, Batch).

Research Computing Technical Staff (part-time) · MIT Office of Research Computing & Data — Nov 2025 – May 2026

- Supported MIT's 3,000+ cluster users on Slurm scheduling, Apptainer containers, and software environments; advised researchers on parallel computing and resource allocation.
- Built tooling for running LLM agents on MIT's supercomputers and wrote the reference recipe for deploying OpenClaw on HPC (Apptainer isolation, SSH tunneling, Lmod).
- TA'd 6 workshops on parallel programming, GPU computing (CUDA), and distributed deep learning.

Machine Learning Intern · Numenta, Menlo Park, CA — Jun – Sep 2025

- Designed a knowledge-preserving method to grow pretrained LLMs from 70M to 1.2B parameters, with distributed multi-GPU training, W&B tracking, and automated evaluation; projected 1.5–2$\times$ compute savings.
- Explored sparse training for efficient LLM inference, reaching $\sim$2$\times$ speedup on CPU and wafer-scale hardware.

Graduate Researcher · MIT Brain & Cognitive Sciences (Boyden Lab, McGovern Institute) — Sep 2021 – Dec 2025

- Published scaling laws for neural-network models of the *C. elegans* nervous system (IEEE SoutheastCon 2024), then released the training data: 11 public datasets ($\sim$900 animals) unified into one open Hugging Face dataset.
- Built end-to-end HPC pipelines for high-dimensional neural data: preprocessing, multimodal alignment, simulation, forecasting.
- Rebuilt light-sheet and diSPIM microscopes for whole-brain imaging (Onshape CAD, optomechanics) and wrote a 35-script analysis pipeline for 1 TB+ recordings.

Bioinformatics & Software Development Intern · Triplet Therapeutics, Cambridge, MA — 2019, 2021

- Deployed neural networks predicting ASO and siRNA knockdown efficacy, improving candidate-selection accuracy by 15% for rare neurological disorders.
- Automated RNA-sequence analysis on AWS EC2 across 10+ concurrent instances, cutting processing time 40%.

Quality Assurance Analyst · Saint Lucia Distillers, Castries, St. Lucia — 2016 – 2017

SELECTED PROJECTS

Octopus Protocol (arXiv 2026, lead author). A coding agent identifies each connected hardware device, then writes, tests, and serves an MCP server for it in about 25 minutes; demonstrated on a Pi-driven robotic arm and cameras.

GrowLab (Sundai Hack, 2026). FLOP-budgeted harness in which an autonomous research agent searches an LLM's parameter-growth schedule; red-teaming our own result traced a 0.913-nat headline to a shared learning rate (corrected: 0.397).

MindMeld (1st place, BrainStorm BCI 2026; team of 6). Real-time 1,024-channel neural decoder on edge hardware: $\sim$0.2 MB model, 94.2% balanced accuracy, <5 ms latency; scored 91.7/100, 22.5 points ahead of second.

Claude Code tooling. 40+ self-written agent skills for evaluation, review, and research workflows, plus *session-merge*, an open-source CLI that repairs fragmented Claude Code sessions.

DriftPak (HardMode 2026, MIT Media Lab; team of 6). AI backpack that sees, hears, speaks, and localizes itself, built from a Raspberry Pi and three ESP32 sensor boards in a 24-hour hackathon.

ChatMerge & Polyglot Mirror. A cross-provider app that merges LLM conversations via embeddings, and a real-time speech-translation mirror with face-tracked captions (2nd most-liked Sundai project, Jan 2026).

PUBLICATIONS

- **Q. Simeon**, J. M. Wei, Y. Fan. "Octopus Protocol: One-Shot Hardware Discovery and Control for AI Agents via Infrastructure-as-Prompts." *arXiv:2605.09055*, 2026.
- **Q. Simeon**. "Recovering Sparse Neural Connectivity from Partial Measurements: A Covariance-Based Approach with Granger-Causality Refinement." *arXiv:2603.18497*, 2026.
- **Q. Simeon**. "*C. elegans* as a Platform for Multimodal Neural Data Integration." M.S. thesis, MIT, 2025.

- **Q. Simeon**, L. Venâncio, M. Skuhersky, A. Nayebi, E. S. Boyden, G. R. Yang. "Scaling Properties for Artificial Neural Network Models of a Small Nervous System." *IEEE SoutheastCon*, 2024.
- **Q. Simeon**, A. Kashyap, K. P. Kording, E. S. Boyden. "Homogenized *C. elegans* Neural Activity and Connectivity Data." *arXiv:2411.12091*, 2024.
- G. Haspel *et al.* (37 authors, incl. **Q. Simeon**). "The Time Is Ripe to Reverse Engineer an Entire Nervous System: Simulating Behavior from Neural Interactions." *arXiv:2308.06578*, 2023.

EDUCATION

M.S., Computational Neuroscience · Massachusetts Institute of Technology 2025
B.S., Computation & Cognition; minor in Statistics & Data Science · MIT · GPA 4.9/5.0 2021

SKILLS

Evaluation	LLM-as-judge, rubric design, inter-annotator agreement (κ, MCC), benchmark and task design, error analysis
Agents & LLMs	Claude Code and coding agents, MCP, multi-agent workflows, Anthropic / OpenAI / Gemini / Bedrock APIs
ML engineering	Python, PyTorch, Hugging Face, distributed multi-GPU training, Weights & Biases, scikit-learn
Infrastructure	Docker, Harbor, AWS (Bedrock, S3, EC2, Lambda, Batch), Slurm/HPC, Apptainer, Linux, Git, CI

HONORS

- 1st Place, Neural Decoding Challenge, BrainStorm BCI Hackathon (Precision Neuroscience) 2026
- Featured in *MIT Spectrum*: "Unlocking AI's Mysteries Using Insights from Brain Research" 2026
- K. Lisa Yang ICoN Graduate Fellowship, MIT 2025 – 2026
- Honorable Mention, Envisioning the Future of Computing Prize, MIT SERC 2025
- Graduate Research Spotlight, *The Bridge* (IEEE-HKN) 2024
- Henry E. Singleton (1940) Fellowship, MIT 2023 – 2024
- Rhodes Scholarship Finalist, Jamaica & Commonwealth Caribbean 2021

TALKS & POSTERS

- Analysis and Modeling of Connectomes, HHMI Janelia, poster and blitz talk 2025
- "*C. elegans* as a Platform for Multimodal Neural Data," NeuroLunch, MIT BCS 2025
- Bernstein Conference on Computational Neuroscience, Frankfurt, poster 2024
- "Scaling Properties for ANN Models of *C. elegans*," Yang Tan Collective, MIT McGovern 2023

TEACHING & COMMUNITY

- Teaching Assistant, *Principles of Neural Computation*, MIT 2023
- Teaching Assistant, *Emergent Computations in Distributed Neural Circuits*, MIT 2023
- Instructor, MIT Global Teaching Labs (Johannesburg, Cardiff) and SPISE programming camp 2019 – 2020
- Tutor, *Fundamentals of Programming* and *Introduction to Neural Computation*, MIT 2018 – 2019
- Builder, Sundai Club weekly AI hackathons 2025 – Present
- Co-President, IEEE-HKN (MIT); Secretary, Theta Chi; Social Chair, MIT Black Student Union 2017 – 2022